

Local Pieces, Global Solutions: A Machine-Checked Sheaf-Cohomological Account of Set-Theoretic Estimation

From Combettes’ feasibility sets and Carlson’s cryptanalytic STE to a mechanized gluing theorem,
with an empirical corpus instantiation — written to be readable without prior cohomology

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Abstract

Set-theoretic estimation (STE) answers an estimation problem by intersecting property sets: an estimate is acceptable exactly when it is consistent with every piece of information [16]. When the hypothesis space is a product over N variables, the feasibility set lives behind an exponential wall, and the standard escape is *locality*: keep small tables over small, overlapping sets of variables. Every local scheme lives or dies on one question — *when does locally consistent partial information determine a global solution?* The mathematics of that question is sheaf theory and Čech cohomology [3, 4], and this paper develops it for STE in machine-checked form (Lean 4 / Mathlib [17, 34]) while assuming no topology of the reader: every term — cover, section, compatible family, gluing, obstruction — is defined from STE primitives, with its constraint-satisfaction meaning stated in one sentence beside it. The main theorem, mechanized sorry-free in the module `Ste.SupportCover`, is the *support-cover gluing theorem*: over any cover of the variables, every compatible family of local sections glues to a global solution *iff* the constraint is generated by pieces whose scopes fit inside the cover — so gluing decides representability over given scopes, and the entire computational difficulty of STE relocates into *computing local sections*, with gluing itself free. Around this centerpiece we prove exact obstruction counts on coupling constraints, rectangle-cover and fooling-set bounds with machine-checked incomparability results, and a reading of the bounded-treewidth algorithmic stack (bucket elimination, junction trees, adaptive consistency) as the theory of when local sections are cheap. We connect the picture to Carlson’s cryptanalytic instantiation of STE [12], whose mechanized counting and reduction theorems we reinterpret sheaf-wise — honestly separating what maps, what is dispatched to in-flight mechanization, and what is speculation. Finally we report a constructive proof-of-work: a verified solver whose pairwise consistency check ran on three historical text corpora, together with the mechanized bridge theorems showing that for frame-structured corpora this pairwise check is a complete feasibility test — the cohomological theory explains *why* the cheap check suffices. Throughout, a strict bright line separates machine-checked results (each cites its Lean identifier) from conjecture, and every heavy cochain construction is quarantined in a clearly marked appendix.

1 Introduction

This paper tells one story three times, at increasing precision: in plain constraint-satisfaction language, in the vocabulary of sheaves built up from STE primitives, and in the named theorems of a Lean 4 mechanization. The story: an STE problem intersects property sets; intersections over

product spaces hide an exponential wall; local reasoning over small variable scopes is the escape; local reasoning is sound and complete exactly when local pieces *glue*; and the gluing question has an exact, machine-checked answer — it succeeds precisely when the constraint is cut out by pieces that fit the chosen scopes. Everything else in the paper — obstruction counts, representation bounds, the algorithmic stack, the cryptanalytic and corpus instantiations — is that one theorem seen from different distances.

Contributions.

1. An *accessible on-ramp* to the sheaf view of STE: every topological word is defined from STE primitives before use (Sections 6–7), with worked examples small enough to check by hand.
2. The *support-cover gluing theorem* (Section 8), mechanized sorry-free (**Ste.SupportCover**): the Čech obstruction of a cover vanishes iff the constraint admits a generating family refining the cover. Corollaries: gluing decides representability over given scopes; the obstruction is a certificate of non-decomposability; and (marked as sketches) a “gluing width” invariant equal to generating arity, provably distinct from treewidth.
3. Machine-checked *quantitative* results (Section 8): exact obstruction counts on coupling constraints, rectangle-cover and fooling-set bounds, and incomparability theorems refuting the slogan “obstruction size = representation blow-up.”
4. An honest *Carlson connection* (Section 9): which parts of the dissertation’s STE cryptanalysis [12] map onto the sheaf picture (with mechanized witnesses), which parts are dispatched to in-flight mechanization, and which do not map.
5. An *empirical instantiation* (Section 10): a verified solver run on three historical corpora, with the mechanized equivalence between its pairwise check and genuine feasibility — and the cohomological explanation of why pairwise suffices for this constraint class.

The bright line. One convention is enforced throughout and is non-negotiable. Every machine-checked claim cites its Lean identifier in **this font**; the named theorem exists, sorry-free, in the project library under continuous integration. Everything not so cited is explicitly labelled: *conjecture*, *sketch*, *dispatched* (an exact statement handed to an in-flight mechanization effort, not yet part of the trusted base), or *speculation*. One refinement: a small number of results were mechanized, during the writing of this paper, on named research branches not yet merged into the main library; these are cited with their branch and commit stated inline, after independent verification that the branch file is sorry-free and compiles, and they are never silently mixed with main-library citations. Adding intuition never relaxes this line; intuition sits beside the citations, not in place of them.

How to read this paper. A reader who knows STE or constraint satisfaction but no algebraic topology can read straight through: Sections 2–4 need no new vocabulary, Section 6 builds the five words the rest uses, and no commutative diagram appears outside the appendix. Asides marked **[For the cohomologist]** and Appendix A carry the cochain-level machinery and can be skipped without losing the thread.

2 Set-theoretic estimation: foundations

STE, as founded by Combettes [16] (with set-membership antecedents in 32), replaces optimization by *feasibility*. An estimation problem is specified by a solution space Ξ , an index set I of pieces of information, and for each $i \in I$ a *property set* $S_i \subseteq \Xi$: the estimates consistent with that piece of information. The answer object is the *feasibility set*

$$T = \bigcap_{i \in I} S_i,$$

and a set-theoretic estimate is any point of T . The mechanization of this core is `Ste.Basic: feasibilitySet`, `mem_feasibilitySet`, the refinement law `feasibilitySet_subset`, partial enforcement `partialFeasibilitySet` with its monotonicity `partialFeasibilitySet_antitone` (more information never enlarges the feasible set), and Combettes’ taxonomy: a formulation is *consistent* if $T \neq \emptyset$, *fair* if the true object lies in T (`Fair`, whence consistency: `feasibilitySet_nonempty_of_fair`), and *ideal* if T is exactly the true object (`Ideal`, `Ideal.eq_of_mem`).

Throughout this paper the hypothesis space is a product: $\Xi = \prod_{v \in V} A_v$, a hypothesis assigning a value to each of N variables, each property set constraining some of them. This is the constraint-satisfaction shape of STE, and it hides a wall. The feasible set lives in a space of size exponential in N , and the project has machine-checked that no representation cleverness can help in general: crisp property sets realize *any* of the 2^N extensional states (`card_allExactStates`, `arbitrary_subset_as_single_property`, in `Ste.DynamicFrame.FiniteSolver`). Any solver that manipulates T extensionally pays 2^N . The rest of the paper is about the principled way around: reason *locally*, and understand exactly when local reasoning is complete.

3 Carlson and the topological turn

Between Combettes’ analytic setting (metric and Hilbert spaces, projections onto convex sets) and this paper’s combinatorial one sits Carlson’s dissertation [12], which recast STE over finite, discrete *topological* spaces and applied it to cryptanalysis. The move matters for us twice over: it is the project’s motivating instantiation of STE outside signal processing, and it is the first place in this lineage where the *topological* character of feasibility — closed sets, compactness, intersection structure — is taken seriously as the load-bearing mathematics. This paper’s sheaf layer is the continuation of exactly that turn: from the topology of the feasible set to the topology of *where information lives*.

3.1 Decryption as STE

Carlson’s recast, mechanized in `Ste.Carlson.Cipher`: the solution space is a finite key space; each observed ciphertext c contributes the property set of keys under which c decrypts to a meaningful message; decryption succeeds when the feasibility set collapses to the true key — Shannon’s unicity point [33] in set-theoretic form. Machine-checked: the cipher-system skeleton (`CipherSystem`, unambiguous decryption `decrypt_unique`), the induced property sets (`keyPropertySet`) and feasible keys (`feasibleKeys`, `mem_feasibleKeys`), fairness of genuine traffic — if the observations really are encryptions under a true key, that key is feasible (`fair_of_genuine`, `feasibleKeys_nonempty_of_genuine`) — unicity as ideality (`Unicity`, `Unicity.eq_of_feasible`), and observation monotonicity: more intercepted traffic can only narrow the key search (`feasibleKeys_antitone`, a special case of `partialFeasibilitySet_antitone`).

The topological well-posedness facts Carlson’s setting needs are mechanized in `Ste.Topology`: closed property sets give a closed feasibility set (`isClosed_feasibilitySet`; discrete case `isClosed_feasibilitySet_of_discrete`), fairness gives the finite intersection property (`FiniteConsistency.of_fair`), and compactness turns finite consistency into consistency outright (`feasibilitySet_nonempty_of_compact`); ideal formulations have closed feasible sets in any T_1 space (`Ideal.isClosed_feasibilitySet`).

3.2 Counting and reduction, machine-checked

Two quantitative pillars of the dissertation are mechanized exactly.

Residual key counting (`Ste.Carlson.Counting`). For a substitution cipher over an alphabet A — keys are permutations of A — an intercepted message constrains the key only on the set T of symbols that actually occur. Carlson’s Lemma 4.1 counts the keys equivalent to the true key: exactly $(|A| - |T|)!$ of them (`card_consistent_keys`, via `card_perm_fixing`; the key space itself has size $|A|!$, `card_substitution_keys`). This is the size of the STE feasible key set after one observation — the residual ambiguity.

Cipher reduction (`Ste.Carlson.Reduction`). On n -bit blocks, every permutation-cipher key (a permutation of bit positions) is realized as a substitution-cipher key (a permutation of the value space) by the induced map `coperm` (`permutation_reduces_to_substitution`), injectively (`coperm_injective`); but for $n \geq 2$ there are strictly more substitution keys than permutation keys (`card_permutation_lt_card_substitution`), so no faithful embedding is onto (`substitution_not_reducible_to_permutation`) — Carlson’s Lemma 5.1/Corollary 5.2 — and permutation–substitution–permutation compositions collapse to a single substitution key (`psp_reduces_to_substitution`, Corollary 5.3; the same reduction is developed in 13, 25).

Section 9 returns to Carlson with the sheaf vocabulary in hand and asks, honestly, which of these constructions are secretly about local sections and gluing.

4 Why sheaves and cohomology are a useful play

The escape from the 2^N wall that every practical system reaches for is *locality*: instead of the monolithic T , keep small tables about small, overlapping sets of variables — per-variable domains, arc-consistent pair tables, junction-tree bags — and reason with those. Every such scheme lives or dies on one question:

If I know everything the problem permits locally — on each small set of variables — and my local answers never contradict each other, do they determine a genuine global solution?

Sometimes yes, sometimes no. When yes, local reasoning is *complete*: the cheap, small tables are a faithful proxy for the exponentially large T . When no, the local tables admit *phantoms* — mutually consistent local answers realized by no global solution — and any margins-only procedure wastes work on them, or worse, reports them. The mathematical bookkeeping for exactly this local-versus-global question is called sheaf theory, and the device that counts the failures is called Čech cohomology. The names do not matter; the payoff does:

- **An exact criterion** for when local reasoning over given scopes is complete (the support–cover theorem, Section 8) — machine-checked, with no structural side conditions on how the scopes overlap.

- **Certificates:** a single stuck family of local answers certifies that *no* decomposition of the constraint respects those scopes; a fooling set certifies that no small union-of-rectangles representation exists (Section 8.5).
- **A precise location for hardness:** the theory proves gluing is never the expensive step — computing genuine local tables is — which turns the bounded-width algorithmic stack into a completeness theorem for local reasoning (Section 8.4).
- **A no-free-lunch theorem:** any invariant computed from the shape of the scopes alone, ignoring the tables, is provably useless (machine-checked; Appendix A.1) — which tells you where *not* to look for speedups.

Where can the local-to-global question even fail? The development contains three distinct axes of locality, and it is worth fixing at the outset that the question is only genuinely hard on one of them.

1. **Splitting the constraint list** (index set I). Partition or arbitrarily cover the constraints, intersect within groups, intersect the results: you always recover exactly T . The map $J \mapsto \text{feas}(J) = \bigcap_{i \in J} S_i$ sends unions of constraint sets to intersections of partial feasible sets (`partialFeasibilitySet_iUnion`), so any cover of the constraint index recovers the global feasible set (`feasibilitySet_eq_iInter_cover`, in `Ste.Sheaf`). *Local-to-global holds here for free*; nothing can obstruct it. (Under the formal-concept reading of `Ste.HyperFrame`, `feas` is a lower polar of the satisfaction context [24] (`partialFeasibilitySet_eq_lowerPolar`), and polars always turn joins into meets.)
2. **Splitting the variable set** (scopes $W \subseteq V$). Keep, for each small scope, the partial assignments the problem permits. Here local-to-global genuinely fails, the failure is precisely variable *coupling*, and this is where the paper lives.
3. **Splitting the evidence corpus** (document sets D , in `Ste.DynamicFrame`). Feasibility over $D \cup E$ splits exactly: feasible on the union iff feasible on each piece *and* the finite, explicitly computable set of constraints active only on the union is satisfied (`crossObstruction`, `mem_feasible_union_iff`, in `Ste.DynamicFrame.PresheafGluing`); a common extension of compatible local worlds exists iff that correction set is empty, and is then unique (`globalExtension_nonempty_iff`, `globalExtension_subsingleton`, `glue`). A local-to-global theorem *with an exact error term*, already closed.

The slogan the mechanization supports: *the constraint side glues for free, the corpus side glues with an exact and computable correction term, and the variable side is where the real local-versus-global phenomenon occurs*. From here on, “local” means local in the variables.

5 Related work

Sheaf-theoretic contextuality. The program closest to this paper is Abramsky–Brandenburger’s sheaf-theoretic treatment of non-locality and contextuality [3], its cohomological refinement [4], the logical reading via paradox [5], contextual semantics across disciplines [2], the quantitative contextual fraction [6], and the studies of when cohomological invariants are complete or have false negatives [1, 14, 15, 30]. The present work differs in two ways: the section presheaf of STE consists of *globally extendable* partial assignments (their empirical models are arbitrary compatible

data), which collapses the per-section obstruction (`amb_extension_always`, Appendix A.2) and moves the interesting content to family-level gluing; and every result here claimed as proven is machine-checked in Lean 4 [17, 34].

Structural constraint satisfaction. The algorithmic layer this paper reinterprets is classical: consistency enforcement [23, 29], constraint networks and directional/adaptive consistency [20], tree clustering [21], bucket elimination [18, 19], with the width theory of Bertelè and Brioschi [9], Robertson and Seymour [31], Arnborg et al. [8], Bodlaender [10], and the homomorphism-side tractability characterization of Grohe [26]. The paper’s contribution to this literature is not a new algorithm but a mechanized exact boundary: which covers admit complete local reasoning at all, independent of acyclicity hypotheses.

Representation lower bounds. The rectangle-cover number and fooling sets are imported from communication complexity [7, 28]; the extension to LP extension complexity [22, 35] is flagged as outlook.

STE lineage. Foundations [16], set-membership estimation [32], and the cryptanalytic line [12, 13, 25, 33].

6 Background: cohomology, built from STE primitives

This section defines every piece of vocabulary the paper uses. Each definition is an ordinary constraint-satisfaction object; the topological name is recorded so that the Lean files and the literature can be read, but nothing later depends on knowing why topologists chose it.

6.1 The five words

Scope, and cover. A *scope* (the Lean files say *context*) is a subset $W \subseteq V$ of the variables. A *cover* is a choice of scopes $U : J \rightarrow \mathcal{P}(V)$ such that every variable appears in at least one ($\forall v \exists j, v \in U_j$). That is all a cover is: **a choice of overlapping variable-scopes to reason locally over**. Three examples recur: the *singleton cover* $\{\{v\} : v \in V\}$ (pure per-variable reasoning); the *bag cover* given by the bags of a junction tree or elimination order; the *trivial cover* $\{V\}$ (no locality at all).

Local section. For a constraint $T \subseteq \prod_v A_v$ and a scope W , a *local section of T over W* is a **partial assignment on W that extends to a full solution of T** — the restriction to W of some member of T . The set of them, $\mathcal{S}_T(W) = \text{localSections } T \ W$, is the image of T under coordinate restriction (`Ste.VariablePresheaf`). Note the convention carefully: a local section is not merely “locally plausible”; global extendability is baked into the definition. (Partial assignments that are only locally consistent — what a propagation algorithm computes — are a weaker object; Section 8.4 names them *pseudo-sections*, and the distinction carries much of the story.) Restriction behaves as it should — restricting in two steps equals restricting in one (`restrict_restrict`, `localSections_restrict`). The topologist’s word for “a scope-indexed family of section sets with well-behaved restriction” is a *presheaf*; that bookkeeping is all the word means here.

Compatible family. Fix a cover U . A *compatible family* is a choice of one local section $s_j \in \mathcal{S}_T(U_j)$ per scope such that any two agree on the overlap of their scopes (`CompatibleFamily`, in `Ste.CechCover`): **local answers that never contradict each other on shared variables**. This is exactly the state of knowledge of a solver that has computed perfect local tables and checked them pairwise.

Gluing. A compatible family *glues* (`GluesCover`) when a single global solution $f \in T$ restricts to s_j on each U_j : **the local pieces can be stitched into one global solution**. When every compatible family over U glues, the *sheaf condition* holds for U (`CechVanishesCover`); in topological notation “ $\check{H}^1$ vanishes for the cover,” and its STE meaning is exactly “**every locally consistent choice is globally realizable.**”

Obstruction. When the sheaf condition fails, the failure can be counted:

$$\text{obs} = \#\{\text{compatible families}\} - \#\{\text{families that glue}\},$$

the number of ways local answers can be mutually consistent yet stitchable into no global solution. The singleton-cover count is `cechObstruction`; the cover-level count is `coverCechObstruction`, and it vanishes whenever the sheaf condition holds (`coverCechObstruction_eq_zero`). Each stuck family is a *phantom*: a profile of partial assignments that every margins-only procedure must entertain although no solution produces it.

Table 1 collects the dictionary.

topological word	STE meaning	Lean name
context / open set	scope: a subset $W \subseteq V$ of variables	—
cover	choice of overlapping scopes hitting every variable	(hypothesis <code>CechVanishesCover</code>) of
local section over W	partial assignment on W extending to a solution	<code>localSections</code>
presheaf	scope-indexed section sets, consistent restriction	<code>restrict_restrict</code>
compatible family	local answers agreeing on shared variables	<code>CompatibleFamily</code>
gluing	stitching the pieces into one global solution	<code>GluesCover</code>
sheaf condition ($\check{H}^1 = 0$)	every compatible family glues	<code>CechVanishesCover</code>
Cech obstruction	count of compatible-but-unstitchable families	<code>cechObstruction</code> , <code>coverCechObstruction</code>

Table 1: The five-word dictionary. Everything in this paper is one of these objects.

Why would a compatible family ever fail to glue? Each piece individually extends to a global solution — that is the definition of a local section. The catch: each piece may extend to a *different* solution, with no single solution extending all of them at once. That one sentence is the entire phenomenon; the next subsection exhibits it in four rows of a truth table.

6.2 The smallest example, worked by hand

All statements in this subsection and the next are proved in Lean, sorry-free (the singleton-cover computations live in `Ste.CechObstruction`); theorem names are cited exactly.

Two Booleans that must agree. Take two Boolean variables x, y and the single constraint “ $x = y$ ”:

$$\Delta = \{f : \{x, y\} \rightarrow \text{Bool} \mid f(x) = f(y)\}$$

(`diagonal`, in `Ste.Sheaf`). It has exactly two solutions, (ff, ff) and (tt, tt) (`diagonal_encard`). Reason locally over the singleton cover: one scope $\{x\}$, one scope $\{y\}$.

Local sections. Over $\{x\}$: does $x = \text{ff}$ extend to a solution? Yes ((ff, ff)). Does $x = \text{tt}$? Yes ((tt, tt)). So both values of x are local sections, and symmetrically for y (`contextSections_diagonal`): every partial view is fully uninformative — each variable, seen alone, can still be anything.

Compatible families. The two scopes do not overlap (`singletonCover_overlap_empty`), so there is nothing to agree on: compatibility is automatic, and a compatible family is just a choice of one value per variable — four families in all (`compatibleFamilies`, `compatibleFamilies_diagonal_encard`):

$$(\text{ff}, \text{ff}) \quad (\text{ff}, \text{tt}) \quad (\text{tt}, \text{ff}) \quad (\text{tt}, \text{tt}).$$

Which glue? A family glues iff, read as a full assignment, it is a solution (`glues_iff_mem`, `gluedFamilies_eq`). Two do (`gluedFamilies_diagonal_encard`). The mixed family $x \mapsto \text{ff}$, $y \mapsto \text{tt}$ is the interesting one: **each half extends to a solution, but to different solutions**, and no single solution extends both. It is compatible and stuck, machine-checked as `diagonal_mixed_compatible_not_glues`. The obstruction count is

$$\text{obs}(\Delta) = 4 - 2 = 2$$

(`diagonal_cechObstruction`). Two phantoms: the two mixed families. This is what a nonzero Čech obstruction *is* — nothing more exotic than the mixed rows of this truth table.

The exact boundary: gluing iff no coupling. Call T *rectangular* if it is a product of per-variable sets, $T = \prod_v P_v$ (`IsRectangle`) — the constraint imposes no coupling, only per-variable filters. Rectangular problems are the fragment where per-variable reasoning is complete: the feasible set of a rectangular family is itself a rectangle (`rectangular_feasibilitySet`) of cardinality $\prod_v |\bigcap_i P_{i,v}|$ (`rectangular_encard`), representable in $\sum_v$ space, and its singleton-cover obstruction is zero (`rectangular_cechObstruction`). The diagonal is provably not rectangular (`diagonal_not_rectangular`); worse, the smallest rectangle containing it is the whole four-point space (`rectangle_superset_diagonal_encard`) — the tightest per-variable view of a coupling carries *zero* information. The singleton-cover story is closed as an iff (`cechVanishes_iff_rectangular`, in `Ste.CouplingLowerBound`):

$$\text{every compatible family glues} \iff T \text{ is a rectangle.}$$

In plain terms: **per-variable local reasoning is exactly as strong as the absence of coupling**. At this cover the obstruction does not merely notice coupling; it characterizes it.

Scaling: n variables that must all agree. The phantom count is not an artifact of small examples. The n -fold coupling $\text{allEqual}(n, \alpha)$ — all n coordinates equal, over an alphabet α (allEqual) — has every per-variable margin full ($\text{contextSections_allEqual}$), exactly as the diagonal did: locally, anything goes. So *all* $|\alpha|^n$ families are compatible ($\text{compatibleFamilies_allEqual_encard}$), while only the $|\alpha|$ constant assignments are solutions (allEqual_encard). The obstruction is exactly $|\alpha|^n - |\alpha|$ ($\text{cechObstruction_allEqual}$), hence at least $2^n - 2$ once $|\alpha| \geq 2$ ($\text{cechObstruction_allEqual_ge}$): exponentially many phantoms while the true feasible set keeps constant size. For $n \geq 2$ the constraint is genuinely non-rectangular ($\text{allEqual_not_rectangular}$, $\text{allEqual_not_cechVanishes}$). A margins-only pipeline on this instance does exponential work distinguishing phantoms from solutions — the 2^N wall of Section 2, seen from the local side.

7 The dictionary, precisely

For reference, this section restates the correspondence at full precision; it adds no new ideas beyond Section 6.

Definition 1 (The section presheaf; mechanized, $\text{Ste.VariablePresheaf}$). *For $T \subseteq \prod_{v \in V} A_v$ and $W \subseteq V$, the local sections are $\mathcal{S}_T(W) = \{f|_W \mid f \in T\}$ (localSections , the image of T under $f \mapsto (v : W) \mapsto f v$). For $W' \subseteq W$ the restriction map $\mathcal{S}_T(W) \rightarrow \mathcal{S}_T(W')$ is coordinate restriction; functoriality is restrict_restrict and $\text{localSections_restrict}$. This makes $W \mapsto \mathcal{S}_T(W)$ a contravariant functor on scopes — a presheaf of sets.*

Definition 2 (Compatibility, gluing, vanishing; mechanized, Ste.CechCover). *For a cover $U : J \rightarrow \mathcal{P}(V)$: a family $(s_j)_{j \in J}$ with $s_j \in \mathcal{S}_T(U_j)$ is compatible (CompatibleFamily) if s_j and s_k agree on $U_j \cap U_k$ for all j, k ; it glues (GluesCover) if some $f \in T$ has $f|_{U_j} = s_j$ for all j ; the constraint satisfies the sheaf condition for U (CechVanishesCover) if every compatible family glues. Glues along a cover are unique (glueCover_unique : two solutions agreeing on every scope of a cover agree everywhere), so the sheaf condition is genuinely existence-and-uniqueness.*

Four machine-checked anchor facts frame the landscape:

- Rectangles glue over every cover, uniquely ($\text{rectangular_cechVanishesCover}$, $\text{rectangular_glueCover_existsUnique}$).
- The trivial cover glues everything ($\text{cechVanishesCover_univ}$): with one scope containing all variables, “local” already means “global.”
- The singleton cover recovers the closed theory of Section 6.2 ($\text{cechVanishesCover_singleton_iff}$), and the diagonal witnesses genuine failure at the cover level ($\text{diagonal_not_cechVanishesCover}$).
- The obstruction count refines the condition ($\text{coverCechObstruction}$, $\text{coverCechObstruction_eq_zero}$; the tautological box-gap form at the singleton cover is $\text{cechObstruction_eq_box_sub_encard}$).

[For the cohomologist.] The correspondence with the standard apparatus: scopes are the opens of the “space” V (discrete), $\mathcal{S}_T$ is a subpresheaf of the product-of-values presheaf, compatibility is the Čech 0-cocycle condition for the cover, gluing is exactness of $F(V) \rightarrow \check{H}^0(U, F)$, and the obstruction count is a cardinality surrogate for the cokernel. The linearized versions — honest cochain complexes with constant

and presheaf coefficients, the degree-one quotients, and the comparison with Abramsky–Mansfield–Barbosa’s γ — are mechanized and discussed in Appendix A.

So gluing depends on the cover, monotonely in spirit: coarse covers glue easily, fine covers glue rarely, and somewhere in between sits a boundary. The next section states the machine-checked theorem that locates the boundary exactly.

8 Main theorems: gluing works exactly when the pieces fit the cover

8.1 The support–cover gluing theorem

One more everyday notion. A constraint S has *support* $\sigma \subseteq V$ if membership in S depends only on the coordinates in σ (**HasSupport**, from **Ste.Support**) — σ is the constraint’s *scope* in the ordinary CSP sense. Say a family (S_i) with scopes σ_i *generates* T if $T = \bigcap_i S_i$, and say its scopes *refine* the cover U if every σ_i fits inside some single scope U_j .

In plain language first: **stitching local answers into global solutions works, for every compatible family, exactly when the constraint can be written as a conjunction of pieces each of which fits inside one scope of the cover.** If the cover’s scopes are big enough to each swallow whole constraints of some generating family, gluing is guaranteed; if no generating family fits, some compatible family is stuck. There is no third case.

Theorem 1 (Support–cover gluing; **mechanized**, **Ste.SupportCover**). *Let $T \subseteq \prod_v A_v$ and let $U : J \rightarrow \mathcal{P}(V)$ be a cover of the variable set.*

- (a) *If T is generated by a family whose scopes refine U , then every compatible family of local sections glues to a global solution — **CechVanishesCover** TU holds (**cechVanishesCover_of_refiningGenerators**).*
- (b) *Conversely, if **CechVanishesCover** TU holds, then $T = \bigcap_j \pi_{U_j}^{-1}(\mathcal{S}_T(U_j))$: T is generated by the cylinder constraints “your restriction to U_j is a local section” (**sectionCylinder**, supported on their scopes by **hasSupport_sectionCylinder**) — a generating family whose scopes are exactly the scopes of U ; indeed gluing is equivalent to this cylinder identity (**cechVanishesCover_iff_eq_iInter_sectionCylinder**).*

Hence the Čech obstruction of the cover U vanishes iff T admits a generating family whose scopes refine U — packaged as “gluing $\iff$ there is a per-scope generating family S_j with **HasSupport** $(S_j)(U_j)$ ” (**cechVanishesCover_iff_exists_scopedGeneration**).

*Proof idea (exposition only; the machine-checked proof is in **Ste.SupportCover**).* (a) Given a compatible family (s_j) , build the only candidate global assignment there is: for each variable v , pick any scope $U_{j(v)}$ containing v and set $f(v) := s_{j(v)}(v)$. Agreement on overlaps makes the choice of scope immaterial, so f restricts to s_j on *every* scope. Why is f a solution? Check the generators one at a time. A generator S_i has its whole scope σ_i inside some single U_{j_0} ; on that scope, f agrees with s_{j_0} , which agrees with the global witness $f_{j_0} \in T$ that makes s_{j_0} a local section. Since S_i only reads coordinates in σ_i , and f matches a known solution there, $f \in S_i$. Every generator passes, so $f \in T$. The single place the hypothesis is used is that each generator’s scope fits inside one scope of the cover — a constraint straddling two scopes could read a mixture of two different witnesses and fail. (Uniqueness of the glue is **glueCover_unique**.)

(b) The cylinders visibly contain T ; conversely an assignment f in all cylinders restricts to a local section on every scope, those restrictions automatically agree on overlaps (they all come from f), gluing produces a solution g agreeing with f on every scope, and since the scopes cover V , $g = f$ (again `glueCover_unique`), so $f \in T$. $\square$

Remark 1 (Sanity anchors, all machine-checked). *At the singleton cover, “a generating family with singleton scopes” means “a rectangle,” and direction (b) specializes to `cechVanishes_iff_rectangular`. Direction (a) contains `rectangular_cechVanishesCover`: singleton scopes refine every cover. The diagonal’s failure is consistent: its minimal scope is the full pair $\{x, y\}$ (`diagonal_support_full`, in *Ste.Support*), contained in no singleton. The trivial-cover success (`cechVanishesCover_univ`) is the case where the single scope swallows all generators. The theorem also discharges, at the cover level, the “residual conjecture” left open in the docstring of *Ste.VariablePresheaf* (gluing holds precisely for the tractable fragments) — provided “tractable fragment” is read as scope-refinement, which by Observation 1 below is emphatically not computational tractability.*

Remark 2 (No structural conditions on the cover). *Readers who know the CSP decomposition literature will notice something missing: classical local-to-global results (tree-clustering, Vorob’ev-style theorems [21, 23]) demand that the cover’s scopes hang together tree-likely (the running intersection property). Theorem 1 imposes nothing of the sort — any cover, any overlap pattern. The reason is the convention of Section 6.1: STE’s local sections are globally extendable by definition. The classical hypotheses earn their keep when the local tables are merely locally consistent rather than genuinely extendable (Section 8.4).*

[For the cohomologist.] In Abramsky–Brandenburger terms: empirical models are arbitrary compatible presheaf data and can be contextual; the presheaf of genuine sections of a global set cannot be, in the per-section sense — the mechanized `amb_extension_always` (Appendix A.2) makes the per-section obstruction identically zero — yet family-level gluing still fails exactly when the cover is too fine for the constraint arity. This is also why no nerve-acyclicity hypothesis appears.

8.2 Effective outcome 1: gluing decides representability

Theorem 1 is an *iff*, and each direction is an effective statement.

Deciding representability over given scopes. “Can T be written as a conjunction of constraints living inside the scopes $U_1, \dots, U_m$?” is, by the theorem, the same question as “does every compatible family over U glue?” — and direction (b) even supplies the canonical decomposition when the answer is yes: the section cylinders (`sectionCylinder`), computable from the local section tables alone. At the singleton cover this specializes to a decision procedure for rectangularity (`cechVanishes_iff_rectangular`); at 1 rectangle it meets the representation bounds below (`rectCoverNumber_eq_one_iff_cechVanishes`).

Certifying non-decomposability. Contrapositively, a *single* stuck compatible family over U certifies that no generating family of T refines U — no decomposition of the constraint respects those scopes, so any pipeline treating the scopes as self-contained is unsound for T . The general contrapositive is mechanized (`exists_scope_refiningNone_of_not_cechVanishesCover`, in *Ste.SupportCover*)¹.

¹Mechanized after the first draft of the underlying exploration memo; the singleton-cover witnesses (`diagonal_not_cechVanishesCover`, `allEqual_not_cechVanishes`) instantiate it: no unary decomposition of these couplings exists.

8.3 Effective outcome 2: where the hardness lives — gluing width is arity, not treewidth

The next two items are *immediate-looking* consequences of the mechanized theorem, but they are not themselves Lean theorems; they are the clearly-marked conjectural harvest.

Definition 3 (Gluing width; not mechanized). $\text{gw}(T)$ is the least w such that the sheaf condition (*CechVanishesCover*) holds for T over some cover U all of whose scopes have at most $w + 1$ variables. By Theorem 1, $\text{gw}(T)$ is exactly (one less than) the minimal arity of a generating constraint family for T : the smallest scopes T decomposes into.

Observation 1 (Gluing width is arity, not treewidth — so vanishing is not tractability; sketch, not mechanized). gw must not be confused with treewidth, and the library already contains the separating example. $\text{allEqual}(n, \alpha)$ is generated by pairwise-equality constraints — scopes of size 2, so $\text{gw} = 1$ — while its primal graph is complete and its induced width is maximal ($n - 1$; prose observation in *Ste. CouplingLowerBound*, not a Lean theorem). More brutally: graph 3-colorability is generated by edge-scoped disequality constraints, so it glues over the edge cover by Theorem 1(a) (this instantiation is prose, not Lean); deciding $T \neq \emptyset$ is still NP-hard. A vanishing gluing obstruction over the natural cover is compatible with full computational hardness. All the hardness has been pushed into computing the objects the theorem quantifies over — the genuine local sections $\mathcal{S}_T(U_j)$ — and none remains in the gluing.

We regard Observation 1 as the honest answer to “is a vanishing obstruction over a good cover an algorithm?” It is half of one: *gluing is the free half; local sections are the expensive half*. The algorithmic payoff below is precisely the theory of when the expensive half is cheap.

8.4 Effective outcome 3: the algorithmic payoff at bounded width

Bag tables are local sections. The treewidth chain (*Ste.Treewidth* through *Ste.JunctionTree*) is built on the **table** of a constraint on a scope σ : the set of restrictions to σ of solutions. Comparing definitions, **table** σ T and **localSections** T σ are the *same construction* — both are the image of T under restriction — so the machine-checked junction-tree theorems are, without change, statements about local sections over the bag cover:²

- *Faithfulness: the bag tables jointly remember the constraint.* A scoped constraint is exactly the preimage of its bag table (`HasSupport.eq_preimage_table`), and a global assignment is a solution iff it restricts into every bag table (`mem_joinConstraint_iff_restrict`, `junctionTables_faithful`). This is Theorem 1(b) in concrete clothing: membership in T is decided scope-by-scope.
- *Cost: local sections are small when scopes are small.* A bag of at most $w + 1$ variables over alphabets of size at most k tables in at most k^{w+1} rows (`table_encard_le_pow`); a width- w elimination order materializes at most $n \cdot k^{w+1}$ rows in total (`bucketEliminate_total_space`, `junctionTree_size_le`; linear-in- n form `junctionTree_size_linear` at the minimal duplicate-free width `junctionWidth`).
- *Computability: bucket elimination computes genuine sections at bounded width.* Projective bucket elimination computes exactly the projection of the solution set onto the remaining variables at each step (`joinConstraint_projectBucketEliminate` and neighbours in

²The one-line identification lemma is dispatched mechanization work (Section 11); the two definitions are syntactically the same image construction.

`Ste.BucketConsistency`) — genuine local sections, not approximations — decides feasibility (`projectBucketEliminate_decides`, `bucketEliminate_decides`), and extracts a solution backtrack-free (`exists_extension_of_mem_projectBucketEliminate`); the fused statement is `projectBucketEliminate_complete_solver` and, for adaptive consistency proper, `decreasingRun_adaptiveSolver` [18–20].

So the mechanized algorithmic layer already computes the genuine local-section tables over an elimination cover, in $O(n \cdot k^{w+1})$ at induced width w (`inducedTreewidth`, `achievesWidth_inducedTreewidth`; the two-sided bridge to Robertson–Seymour treewidth [31] is `treewidth_primalGraph_le` and `inducedTreewidth_le_treewidth_sub_one`; nonserial dynamic-programming ancestry in 9). Combined with Theorem 1(a) — gluing is free once the generators fit the bags — this is a complete local-to-global pipeline for bounded-width instances: $O(n \cdot k^{w+1})$ preprocessing replaces 2^N enumeration, and bag-local queries read off the section tables soundly by faithfulness.

Consistency algorithms compute pseudo-sections. Local-consistency algorithms [23, 29] compute something weaker than genuine local sections: tables closed under local checks, with no guarantee their entries extend to global solutions. Call these *pseudo-sections*. The gap between pseudo and genuine is exactly where the hardness displaced by Observation 1 ends up living, and the mechanized tractability results are precisely theorems that *pseudo equals genuine* under structural hypotheses: on a chain, arc consistency puts every surviving value on a global solution (`arcConsistent_backtrackFree`, `arcConsistent_chain_solvable`); on a rooted tree, directional arc consistency does the same (`treeArcConsistent_solvable`, `Ste.ConsistencyTree`); at induced width w , directional consistency of the buckets does the same, and bucket elimination establishes it while preserving solutions (`directionalConsistent_solvable` in `Ste.AdaptiveConsistency`; `Ste.BucketConsistency`).

	pseudo-sections (k -consistent tables)	genuine sections (<code>localSections</code>)
cheap to compute	yes (local closure)	only at bounded width
entries globally extendable	not guaranteed	always, by definition (cf. <code>amb_extension_always</code>)
family gluing	fails generically	iff generators fit the cover (Thm. 1)
where hardness sits	pseudo $\neq$ genuine gap	computing $\mathcal{S}_T$

The conjectural summary (a reading, not a theorem): *the cohomology-shaped content of STE hardness is the pseudo/genuine gap, and every tractability theorem in the library is a statement that the gap closes over a structured cover.*

8.5 Effective outcome 4: representation bounds and certificates

A tempting slogan — “the bigger the obstruction, the bigger the forced representation blow-up” — is settled *negatively*, and machine-checked, in `Ste.RepresentationBounds`. The measuring stick is the *rectangle-cover number* $\rho(T)$ (`rectCoverNumber`): the least number of no-coupling (rectangular) constraints whose *union* is exactly T — “how many coupling-free alternatives describe T ?” This is the STE analogue of the nondeterministic communication-complexity cover number [7, 28], with the LP-extension-complexity lineage [22, 35] as outlook. Machine-checked:

- *Bounds.* A *fooling set* — solutions no two of which can be mixed coordinate-wise and stay solutions — forces one rectangle each: $\text{fool}(T) \leq \rho(T) \leq |T|$ (`foolingNumber_le_rectCoverNumber`, `IsFoolingSet.encard_le_of_hasRectCover`, `rectCoverNumber_le_encard`); the mixing engine is `IsRectangle.mix_mem`.
- *The bottom of the scale is the gluing condition:* $\rho(T) = 1 \iff$ the singleton-cover obstruction vanishes (`rectCoverNumber_eq_one_iff_cechVanishes`, `one_lt_rectCoverNumber_iff`).
- *Exact values on the coupling:* $\rho(\text{allEqual}(n, \alpha)) = |\alpha|$ (`rectCoverNumber_allEqual`; fooling set `isFoolingSet_allEqual`, `foolingNumber_allEqual`).
- *No law connects obstruction size to ρ* (`not_forall_cechObstruction_le_rectCoverNumber`, `not_forall_rectCoverNumber_le_cechObstruction`): rectangles give $\text{obs} = 0 < 1 = \rho$; `allEqual(3, Bool)` gives $\rho = 2 < 6 = \text{obs}$ (`rectCoverNumber_lt_cechObstruction_allEqual`); on the two-variable diagonal the two coincide (`diagonal_cechObstruction_eq_rectCoverNumber`) — a coincidence of the smallest example, not a law.

Two representation grammars are in play: ρ measures $\cup$ -of-boxes (DNF-like) representations; the junction-tree machinery measures $\cap$ -of-cylinders (CNF-over-bags) representations. The machine-checked triple $(\text{obs}, \rho, \text{width})$ on `allEqual` shows all three diverge: obstruction exponential, ρ constant, width maximal.

Conjecture 1 (ρ vs. width incomparability, other direction; not mechanized). *A width-1 chain of disequality constraints over an alphabet of size $k \geq 3$ on n variables has feasible set of size $k(k-1)^{n-1}$ and, we conjecture, $\rho = 2^{\Omega(n)}$ (a fooling-set construction along alternating assignments looks plausible). If so, ρ and treewidth are incomparable in both directions: ρ answers “how many boxes,” width answers “how large a bag.”*

Practitioner’s reading: fooling sets are cheap-to-exhibit certificates that no small union-of-boxes representation exists; stuck compatible families over a proposed cover (Section 8.2) are cheap-to-exhibit certificates that no generating family respects those bags. Both certificate kinds are mechanized at the definition level and computed exactly on the coupling family. Cheap NO-certificates steer the expensive machinery: test a candidate bag cover with a small stuck-family search before paying for elimination.

9 Carlson connection points

Does Carlson’s cryptanalytic STE [12] map onto the sheaf picture? The honest answer is: *partially, and the partial map is instructive.* We separate three tiers.

9.1 What maps, with mechanized witnesses

Observation accumulation is the constraint axis. Carlson’s “more intercepted traffic narrows the key search” (`feasibleKeys_antitone`) is an instance of the axis that glues for free (Section 4, item 1; `partialFeasibilitySet_antitone`, `feasibilitySet_eq_iInter_cover`): observations index property sets, and covers of the *observation* index can never obstruct. The dissertation’s accumulation arguments live entirely on this axis, which is why they need no gluing hypotheses.

Residual key counting is fiber counting. Carlson’s Lemma 4.1 (`card_consistent_keys`) counts the permutations agreeing with the true key σ on the observed symbol set T : $(|A| - |T|)!$. In the paper’s vocabulary, a key’s action on T is a partial assignment on the scope T (variables =

alphabet symbols, values = alphabet symbols), and the count is the number of *global witnesses of one local section* — the fiber of the restriction map over $\sigma|_T$. Unicity (**Unicity**) is the statement that accumulated observations shrink every such fiber to one point. This reading is faithful to the mechanized statement as it stands; its formal restatement through the restriction map used by `localSections` is now itself mechanized on a research branch (Section 9.2).

9.2 The scoped-observation theorems (mechanized on a research branch)

The genuinely sheaf-shaped content of the substitution cipher is this: *an observed ciphertext constrains the key only on the symbols that occur in it* — an observation constraint is a *scoped* constraint in the sense of `HasSupport`, with scope the ciphertext’s symbol set (on the decryption-side parametrization of keys). Meanwhile the requirement that the key be a *permutation* — injectivity across all symbols — is a global *coupling*, the cryptanalytic cousin of `allEqual`’s all-different flavor, and it is provably non-rectangular. So Carlson’s instance factors exactly along this paper’s main axis: scoped evidence (gluable, by Theorem 1(a), over any cover the symbol sets refine) against one global coupling (the permutation constraint, which no proper cover swallows).

These statements were dispatched during the writing of this paper and have been **mechanized, sorry-free, on the research branch** `research/carlson-sheaf-bridge` (commit 4ae7425, module `Ste.Carlson.SheafBridge`; independently verified to contain no `sorry` and to compile against the project — axioms limited to the standard three). The branch is *not yet merged* into the main library, so per the bright line these four results are cited with the branch stated and are kept out of the main-library tally: `hasSupport_decConstraint` (the decryption-side observation constraint `decConstraint` has support exactly the ciphertext’s symbol set); `mem_keyPropertySet_iff_decConstraint` (it is Carlson’s `keyPropertySet` of the substitution `subCipher`, read through key inversion); `bijKeys_not_rectangular` (the permutation coupling `bijKeys` is not rectangular over any nontrivial alphabet); and `card_restrict_fiber` (Carlson’s Lemma 4.1 restated through the restriction map used by `localSections`: the permutations restricting on T to a given local section number $(|A| - |T|)!$, derived from `card_consistent_keys`). With these, the fiber-counting reading of Section 9 is a theorem, not an analogy — modulo the merge.

9.3 What does not map

Two honest negatives. First, Carlson’s hypothesis space is a key space, not a product of variables; the sheaf reading above requires re-embedding keys into the function space $A \rightarrow A$ and carrying the permutation constraint separately. Nothing in the dissertation uses cover-locality on the key’s coordinates; the productive structure there is *class-level reduction* — embedding one cipher class’s keys in another’s (`permutation_reduces_to_substitution`, `coperm_injective`, `substitution_not_reducible_to_permutation`, `psp_reduces_to_substitution`) — which is group theory, not gluing, and we see no gluing-theoretic content in it. Second, the dissertation’s information-theoretic layer (unicity distances, entropy of meaningful text) has no mechanized counterpart in this project and no sheaf reading we would defend; connecting the obstruction counts of Section 8.5 to unicity-distance estimates is *speculation* and is recorded as such in Section 11.

10 Empirical analysis and constructive proof-of-work

The project’s empirical layer runs a *verified* STE solver on frame corpora extracted from historical texts. This section reports what ran, exactly what the solver computes, and what the cohomological

theory adds — with the pipeline’s trust boundary stated plainly.

10.1 The frame-corpus constraint class and the verified solver

An empirical corpus is a *frame* family: each author a fixes a partial assignment $\text{frame } a : V \rightarrow \text{Option } W$ of values to variables (silence = none), and the property set of a is the set of total assignments agreeing with a wherever a speaks (`frameConstraint`, in `Ste.FiniteInstance`). For this class the feasibility question collapses to a decidable pairwise check, and the collapse is machine-checked: the corpus is jointly satisfiable iff no two authors assign the same variable different values (`Consistent`, with the bridge `nonempty_iff_consistent` and the verdict form `feasibilitySet_eq_empty_iff`); the quantitative refinement counts distinct values per variable (`disagreementDegree`, characterized by `agree_iff_degree_le_one`).

The executable (`lean/TidesExe.lean`) parses a `frames.json` at runtime and evaluates exactly these verified definitions — `STE.Consistent` and `STE.disagreementDegree` — on the parsed frames. **To be precise about what ran: the solver’s algorithm is the pairwise consistency check of `Ste.FiniteInstance`; it imports no cohomology module, and no Čech computation executed on any corpus.** The trust boundary is the JSON parser and Lean’s evaluator; the aggregation logic is the verified one.

10.2 The runs

Three corpora, one pipeline (LLM extraction of frames from period texts; the verdicts computed by the verified layer):

- *Tides* (`Ste.TidesCorpus`, `Ste.TidesComputable`): pre-Newtonian authors on the cause of the tides, canonicalized to a 7-variable schema. Verdict: globally *inconsistent*, and pervasively so — the five-author “attraction” camp is internally consistent on the moon’s role but not overall (`tides_inconsistent`, `attraction_camp_moon_consistent`); the kernel-computed verdict (`feasible_eq_empty_computed`) agrees with the hand-proved one, and the corpus is definitionally an instance of the general frame class (`constraint_eq_frameConstraint`).
- *Federalist* (86 essays; 27): 10 variables; inconsistent but *narrowly* — 7 of 10 variables unanimous where spoken, exactly 3 conflict sites carrying the whole obstruction.
- *Ratification, two camps* (the 86 Federalist essays plus 73 Anti-Federalist essays; 159 documents): the consistent core shatters — all 10 variables become conflict sites. The contrast $3 \rightarrow 10$ is the sharp empirical signature of adding an opposing camp.

10.3 What the cohomology adds: why the cheap check is the right one

Here is the honest division of labor. The solver ran a *pairwise* check. The theory of this paper explains *why pairwise is enough for this constraint class* — why no higher-order gluing failure can hide in a frame corpus.

The plain argument: each author’s `frameConstraint` is, by construction, a conjunction of *unary* constraints (“variable v has value x ,” one per spoken variable — read the definition: $\forall v x, \text{spoken} \Rightarrow \text{agreement}$). Unary scopes refine *every* cover, so by the mechanized Theorem 1(a) (`cechVanishesCover_of_refiningGenerators`) the feasible set of a frame corpus glues over every cover of the variables; equivalently, the corpus feasible set is a rectangle, the class for which per-variable and pairwise reasoning is complete (`cechVanishes_iff_rectangular`, `rectangular_cechVanishesCover`). There are no phantoms to fear: locally consistent readings of a frame corpus

are always globally realizable, which is exactly why the decidable surrogate `Consistent` can coincide with genuine feasibility (`nonempty_iff_consistent` — itself machine-checked, independent of any cohomology). The *formal* packaging of the rectangularity argument (“`frameConstraint` is rectangular, hence the corpus feasible set is, hence it Čech-vanishes for every cover”) is exact, Lean-ready, and dispatched (Obligation 2); until it lands, the machine-checked pillars are the two endpoints already cited — `nonempty_iff_consistent` on the solver side and Theorem 1 on the theory side — and the connecting rectangularity lemma is the marked gap.

Two consequences worth stating plainly. First, the empirical runs are a genuine, if structurally easy, instance of the paper’s subject: a local-reasoning solver whose completeness is exactly a gluing fact. Second, the interesting failure mode — coupled constraints, where pairwise reasoning is *not* complete (Section 6.2) — does not arise for frame corpora, but will arise the moment extraction produces *relational* claims (“ v_1 and v_2 stand in relation R ”) rather than variable-value assignments. A solver for that regime, carrying its gluing certificate explicitly (compute bag-local sections, check compatibility, invoke Theorem 1 for soundness), is the natural constructive next step; **no such cohomology-labelled solver exists yet**, and we flag it as future work rather than claim it.

11 Conclusion and future work

11.1 What is machine-checked, in one paragraph

The constraint side of STE glues for free (`feasibilitySet_eq_iInter_cover`); the corpus side glues with an exact computable correction (`mem_feasible_union_iff`); on the variable side, per-variable gluing characterizes the absence of coupling exactly (`cechVanishes_iff_rectangular`), coupling-free constraints glue over every cover with unique glues (`rectangular_glueCover_existsUnique`), couplings genuinely fail (`diagonal_not_cechVanishesCover`) with exactly computed, exponentially growing phantom counts (`cechObstruction_allEqual`, `cechObstruction_allEqual_ge`), and the boundary is exact: the support-cover gluing theorem (Theorem 1, `Ste.SupportCover`) with its certificate contrapositive (`exists_scope_refiningNone_of_not_cechVanishesCover`). Cover-shape-only invariants are provably blind (`cechH1_subsingleton`, Appendix A.1) while the section-aware linear obstruction fires over every nontrivial ring (`mixedFamily_not_linearlyGlues`) even though the per-section AMB obstruction cannot (`amb_extension_always`). The rectangle-cover number is sandwiched, characterized at 1 by gluing, computed on the coupling, and provably incomparable to the obstruction magnitude (`not_forall_cechObstruction_le_rectCoverNumber` and mirror). The bounded-width stack — tables, bucket elimination, junction trees, adaptive consistency — is mechanized with faithfulness, $O(n \cdot k^{w+1})$ size, and a fused complete-solver statement (`projectBucketEliminate_complete_solver`). Carlson’s STE recast, counting lemma, and reduction chain are mechanized (`Ste.Carlson`); the frame-corpus solver equivalence is mechanized (`nonempty_iff_consistent`) and ran on three corpora.

11.2 What remains conjectural

The gluing-width invariant and its arity reading (Definition 3, Observation 1); the pseudo/genuine-gap reading of tractability (Section 8.4); the ρ -vs-width incomparability conjecture; the bag-local linear relaxation with exactness controlled by the twisted invariant (Appendix A.2); and any connection between obstruction counts and unicity distance (Section 9) — explicitly speculation.

11.3 Dispatched proof obligations

The following exact obligations are, at the time of writing, dispatched to in-flight Lean mechanization efforts on named research branches. *None of them is cited as proven anywhere in this paper*; when they land sorry-free, the noted sections upgrade.

Proof Obligation 1 (Carlson sheaf bridge; branch `research/carlson-sheaf-bridge` — **discharged**, awaiting merge). (a) *The decryption-side observation constraint of the substitution cipher has `HasSupport` equal to the symbol set of the ciphertext.* (b) *Its equivalence with `keyPropertySet` of the substitution `CipherSystem` via key inversion.* (c) *The permutation coupling $\{g : A \rightarrow A \mid g \text{ bijective}\}$ is not rectangular over a nontrivial alphabet.* (d) *Carlson’s Lemma 4.1 restated through the restriction map (from `card_consistent_keys`).* Status: all four items mechanized sorry-free on the branch at commit `4ae7425` (Section 9.2); they enter the main-library tally on merge.

Proof Obligation 2 (Frame rectangularity; branch `research/frame-rectangular`). (a) *IsRectangle of each author’s `frameConstraint`;* (b) *IsRectangle of the corpus feasible set `feasibilitySet(frameConstraint)`;* (c) *hence `CechVanishesCover` for every cover of the variables;* (d) *optionally, the unary-generation form feeding `cechVanishesCover_of_refiningGenerators`.* Upgrades Section 10 from prose argument to cited theorem.

Proof Obligation 3 (Near-term backlog; branch `research/mechanize-conjectures`, in flight independently of this paper). *The `table = localSections` identification; a pair-cover gluing statement for `allEqual` and a Lean-level gluing-width definition; and the second-tier representation bounds.* Upgrades Sections 8.3 and 8.4.

11.4 Further directions

A pseudo-section presheaf as a first-class object, with the unifying vanishing theorem “running-intersection cover + directional consistency $\Rightarrow$ pseudo = genuine” subsuming the chain, tree, and adaptive cases; the variable-side analogue of the corpus axis’ exact correction term (splice two partial solutions, collect straddling constraints), then its k -piece iteration; a bag-local must/may/cannot query theorem over junction tables; the twisted degree-one theory (Appendix A.2) toward exactness guarantees for linear relaxations in the style of Abramsky et al. [6]; and the relational-claims solver of Section 10 — the first solver that would carry its gluing certificate explicitly. Pleasingly for a paper about cohomology, the content of its main theorem is that in set-theoretic estimation the cohomology, properly aimed, is exactly the bookkeeping of who is allowed to talk to whom — and that bookkeeping is machine-checked.

A For the cohomologist: what the coefficients see

This appendix is the paper’s heavy-machinery annex. Nothing in the main text depends on it beyond the two plain-language morals already stated there: cover-shape-only invariants are provably blind (A.1), and a section-aware linear invariant fires one degree lower than the AMB program would suggest (A.2). The Abramsky–Brandenburger program grades gluing failure cohomologically by linearizing the section presheaf [3, 4]; the library mechanizes two coefficient regimes, and the contrast between them is, in our view, the single most instructive machine-checked fact in this layer.

A.1 Constant coefficients see nothing — a machine-checked no-free-lunch

`Ste.CechComplex` builds the constant-coefficient cochain complex on the full nerve of a cover index — modules $C^0 = \iota \rightarrow M$, $C^1 = \iota \rightarrow \iota \rightarrow M$, ... with the standard alternating coboundaries (`cechD0`, `cechD1`, `cechD2`), the complex identities (`cechComplex_d_comp_d`, `cechComplex_d2_comp_d1`), and honest quotient modules $\check{H}^1, \check{H}^2$ (`cechH1`, `cechH2`). The vanishing theorem (`cechH1_subsingleton`, `cechH2_subsingleton`) proves $\check{H}^1 = \check{H}^2 = 0$ over any nonempty index type: the full nerve is a simplex, and a simplex is acyclic; $\check{H}^0$ is exactly the coefficients (`cechH0EquivCoeff`).

The computational moral: *any invariant computed from the combinatorics of the cover alone — ignoring the section data — is useless for detecting coupling*; the machine checks that it is identically zero. Whatever a “cohomological speedup” means, its input must include the tables of local sections — exactly the data the CSP algorithms of Section 8.4 already manipulate. This is consistent with the contextuality literature, where the interesting classes live in relative cohomology with *presheaf* coefficients [4, 14].

A.2 Twisted coefficients see the coupling — one degree lower than expected

`Ste.TwistedCech` builds the honest twisted complex: coefficients in the linearized section presheaf $F_R(W) = (\mathcal{S}_T(W)) \rightarrow_f R$ (free R -module on genuine local sections, `twistedCoeff`), restriction by pushforward along section restriction (`sectionRes`, `twistedRes`, functorial by `twistedRes_twistedRes`), coboundaries `twistedD0`, `twistedD1` with $d^1 \circ d^0 = 0$ (`twisted_d1_comp_d0`), and $\check{H}^0$ as `twistedH0`. Two machine-checked results locate the STE coupling precisely.

1. *The AMB per-section obstruction degenerates in STE* (`amb_extension_always`). Abramsky–Mansfield–Barbosa attach to a single local section s_1 a class $\gamma(s_1) \in \check{H}^1$ vanishing iff s_1 extends to a compatible family of the linearized presheaf [4, Prop. 4.2]. The mechanization proves that in STE this extension condition holds for *every* section of *every* constraint over *every* cover — because `localSections` only ever contains restrictions of global solutions, the extension witness is free. The $\check{H}^1$ -graded per-section obstruction is structurally blind here: STE’s section presheaf bakes global extendability into its stalks. (The long exact sequence and the literal connecting map γ are *not* mechanized; what is proven is the extension condition AMB prove equivalent to $\gamma = 0$.)
2. *The obstruction that fires lives in the cokernel of $F_R(V) \rightarrow \check{H}^0(U, F_R)$* (`mixedFamily_not_linearlyGlues`, `diagonal_mixed_cocycle_not_in_globalRange`). A compatible family’s point-mass cochain is a twisted 0-cocycle (`familyCochain_mem_twistedH0`); the family *glues R -linearly* (`LinearlyGlues`) iff that cocycle lifts to a formal R -combination of global solutions (`linearlyGlues_iff_mem_range`). For the two-bit coupling and its singleton cover, the mixed family fails to glue linearly over *every nontrivial commutative ring* — negative and rational coefficients included. This is strictly stronger than the set-level failure (`diagonal_gluing_fails`): in the AMB world every no-signalling model has a global section over the signed reals [3], but the STE coupling is not even signed-linearly explainable by global data. Set-level gluing always implies linear gluing (`GluesCover.linearlyGlues`), and rectangles glue linearly over every cover (`rectangular_linearlyGlues`), so nonmembership is a genuine obstruction invariant (`diagonal_twisted_obstruction_detects`).

Whether the twisted $\check{H}^1$ (carried as the submodule pair `twistedCoboundaries1` $\leq$ `ker twistedD1`, with triviality expressed as `TwistedH1Trivial`; the literal quotient type is blocked by

a typeclass diamond, documented in the file) vanishes for pairwise-disjoint covers is stated as *open* in the source and remains open here. It is the first stepping stone toward exactness guarantees for bag-local linear relaxations — signed/affine analogues of the k -consistency hierarchy, cousins of the contextual-fraction linear programs of Abramsky et al. [6] — with the standing caution from the literature [1, 14, 30] that invariants of this style can have false negatives; the mechanized `amb_extension_always` is precisely such a false-negative theorem.

[For the cohomologist.] On the corpus axis the presheaf is separated (identical ambient hypothesis on overlaps), so its theory closes at $\check{H}^0$; a Mayer–Vietoris organization of iterated corpus splits (standard machinery, e.g. 11) would carry correction terms that are precisely the `crossObstruction` sets, and separatedness makes us expect immediate degeneration — a polite way of saying the corpus axis needs no cohomology beyond what is mechanized. A bi-indexed presheaf on $(\text{corpora}) \times (\text{scopes})$, corpus direction separated, variable direction carrying the twisted theory, is (frankly speculative) outlook.

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